

Exercises 59-62 are based on questions found in the book Symbolic Logic by Lewis Carroll.

59) Let $P(x)$, $Q(x)$ and $R(x)$ be the statements " x is a professor," " x is ignorant," and " x is vain," respectively. Express each of these statements using quantifiers; logical connectives; and $P(x)$, $Q(x)$ and $R(x)$, where the domain consists of all people.

a) No professors are ignorant: $\neg \exists x (P(x) \wedge Q(x))$

b) All ignorant people are vain: $\forall x (Q(x) \Rightarrow R(x))$

c) No professors are vain: $\neg \exists x (P(x) \wedge R(x))$

d) Does (c) follow from (a) and (b)?

Let $\neg \exists x (P(x) \wedge Q(x))$ be True. By De Morgan's law of quantifiers, $\forall x \neg (P(x) \wedge Q(x))$ is True. By 1st De Morgan's law, $\forall x (\neg P(x) \vee \neg Q(x))$ is True. By defn. of universal quantifier, for all elements a in the domain, $\neg P(a) \vee \neg Q(a)$ is True. By defn. of OR, at least one of $\neg P(a)$ or $\neg Q(a)$ is True.

Let $\forall x (Q(x) \Rightarrow R(x))$ be True. By conditional disjunction law, $\forall x (\neg Q(x) \vee R(x))$ is True. By defn. of universal quantifier, for all elements a in the domain, $\neg Q(a) \vee R(a)$ is True. By defn. of OR, at least one of $\neg Q(a)$ or $R(a)$ is True.

$\neg \exists x (P(x) \wedge R(x)) \equiv \forall x \neg (P(x) \wedge R(x))$ (De Morgan's law of quantifiers)
 $\equiv \forall x (\neg P(x) \vee \neg R(x))$ (1st De Morgan's law)

$\therefore$ We need to show that for all elements a in the domain, $\neg P(a) \vee \neg R(a)$ is True in order to prove that (c) follows from (a) and (b).

It's easy to see that there can exist an element b in the domain where $\neg P(b)$ is False, $\neg Q(b)$ is True and $R(b)$ is True. $\therefore$ By defn. of negation, $\neg R(b)$ is False. By defn. of OR, $\neg P(b) \vee \neg R(b)$ is False.